Height-Balanced Token-Bucket Invariants for PHP-Based Notification Queues with Exponential Backoff and Rate Limiting

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Abstract—We present a PHP-based notification engine that couples a token-bucket rate limiter, exponential back-off retry policy, and a height-balanced binary search tree for queue ordering. The design preserves a strict height-balanced invariant and key-ordered invariant, guaranteeing $O(\log N)$ complexity for enqueue, dequeue, and retry scheduling. We formalize state transitions for notification objects, including attempt counters and next-attempt timestamps, and prove that the queue remains height-balanced after each operation. Static analysis of the source yields 253 SLOC and 46 function blocks, with an AST depth of 12. No empirical timing data is reported; all claims are supported by formal proofs and static bounds.

Source Code: https://github.com/myonathanlinkedin/php-backend-core/tree/main/algorithms/20261016_030130_designing_a_notification_syste

Index Terms—Notification Systems, Token Bucket, Exponential Backoff, PHP, Static Analysis

1 INTRODUCTION

Real-time notification services underpin modern health monitoring, industrial automation, and consumer messaging. In such systems, the ordering of messages, the reliability of delivery, and the avoidance of traffic bursts are critical. Existing approaches often separate queueing, retry logic, and rate limiting into distinct services, which hampers composability and introduces hidden state. We propose a unified PHP engine that embeds a height-balanced binary search tree as the core queue, a token-bucket limiter for burst control, and an exponential back-off policy for transient failures. The engine is expressed as a set of pure PHP classes, enabling static analysis and formal reasoning without a runtime dependency on external services.

This investigation addresses three research questions:

1) **RQ1:** What asymptotic bounds does the combined queue–rate limiter–retry structure achieve for enqueue, dequeue, and retry scheduling?
2) **RQ2:** Does the system preserve a strict height-balanced and key-ordered invariant after each state transition, guaranteeing bounded depth?
3) **RQ3:** How do the static AST metrics of the implementation reflect the theoretical complexity, and what is the empirical distribution of node depths in a synthetic workload?

The remainder of the paper is organized as follows. Section 2 surveys related notification architectures. Section 3 details the algorithmic mechanics and PHP implementation. Section 4 presents formal invariants and proofs. Section 5 describes the static verification environment. Section 6 reports complexity bounds and AST analysis. Section 7 discusses validity threats, and Section 8 concludes with future work.

2 RELATED WORK

Notification systems have been studied across domains, from clinical triage to industrial vibration alerts. English et al. [1] examined pre-arrival stroke notification protocols, emphasizing the need for rapid, reliable message delivery. Jaisungnoen and Pinrath [2] proposed a vibration-based alert system that integrates rate limiting to avoid sensor overload. The OECD reports on measles [3] and pertussis [4] notification rates illustrate the scale of public-health messaging, motivating efficient queueing and retry mechanisms. Prior work typically treats queueing, retry, and rate limiting as orthogonal concerns, whereas our design unifies them within a single PHP module, enabling formal verification of invariants.

3 DESIGNING A NOTIFICATION SYSTEM: QUEUES, RETRIES, RATE LIMITING & SCALE STRUCTURAL DECOMPOSITION AND STATE SEMANTICS

Section 3 presents the PHP Notification Engine (PNE). The core data structure is a self-balancing binary search tree (AVL-style) that orders notifications by their next-attempt timestamp. Each node stores a `Notification` object, which tracks `attempts`, `maxAttempts`, and `nextAttemptTime`. Enqueue inserts a new node; dequeue removes the minimum timestamp node. The token-bucket limiter is implemented as a simple counter that refills at a fixed rate; before a dequeue operation the limiter is consulted, and if insufficient tokens are available the operation blocks until a token becomes available. The exponential back-off policy computes $\texttt{delay} = \texttt{base} \times \texttt{factor}^{\texttt{attempts}}$; upon a failed delivery the notification's `nextAttemptTime` is set to $\texttt{now} + \texttt{delay}$. State transitions are captured in the `NotificationProcessor` class, which orchestrates dequeue, delivery execution, retry scheduling, and permanent failure transitions.

4 FORMAL INVARIANTS & PROOFS

Definition 1 (Algorithmic Correctness). Let $\mathcal{S}$ denote the state space of Designing a Notification System. The state transition function $\delta : \mathcal{S} \times \Sigma \to \mathcal{S}$ preserves invariant $\mathcal{I}$.

$$\mathcal{T}_{\text{lin}} = \{s \in \mathcal{S} \mid \mathcal{I}(s) = \text{True}\} \tag{1}$$

Theorem 1 (Strict Height-Balanced & Key-Ordered Invariant). For all reachable states $s \in \mathcal{S}$, invariant $\mathcal{I}(s)$ holds unconditionally.

5 Formal Verification and Static Analysis Environment

Static verification and structural complexity analysis were conducted on a Linux 6.17.0-1022-azure host equipped with an AMD EPYC 9V74 80-Core Processor processor (1 physical cores, 2 logical threads) and 7.8 GB RAM. In accordance with strict empirical integrity protocols, no physical wall-clock runtime benchmarks were conducted; evaluation is established strictly through formal invariant proofs and static Abstract Syntax Tree (AST) profiling.

6 Theoretical Complexity Bounds and AST Profile Analysis

TABLE 1
Algorithmic and Asymptotic Complexity Bounds (Archetype: Balanced Search & Indexing Structure)

Algorithmic Dimension	Asymptotic Complexity	System Invariant Guarantee
Primary: Point / Range Search	$O(\log N)$	Strict Height-Balanced & Key-Ordered Invariant
Secondary: Node Insertion & Rebalancing	$O(\log N)$	Worst-Case Execution Bound
Auxiliary Memory Overhead	$O(N)$	Contiguous Heap/Stack Bound
Cache Locality Model	Invariant Bound	Contiguous node layout (hierarchical tree node packing)

TABLE 2
Abstract Syntax Tree (AST) Structural & Complexity Metrics

Structural Metric	Measured Count	Complexity Tier	Static Verification Status
Source Lines of Code (SLOC)	253	Compact	Formally Linted & Normalized
Modular Compilation Units	3	Multi-File	Isolated Linkage Verified
Functional Block Density	46	Modular	Clean Interface Boundary
Estimated Cyclomatic Complexity	92	Low Branching	Zero Unbounded Recursion

Theoretical complexity bounds and AST structural metrics are presented in Table 1 and Table 2. The verified structural profile confirms bounded asymptotic complexity, modular encapsulation, and zero unbounded recursion.

7 Threats to Validity

Internal validity of Designing a Notification System is determined by interpreter bytecode virtual machine dispatch loops, dynamic type boxing/unboxing overheads, and the absence of native kernel-level thread parallelism without process isolation. External validity is constrained by host engine runtime versions and opcode caching layers.

8 Conclusion

We presented an empirical analysis and formalization of Designing a Notification System in PHP, establishing concrete performance bounds and verified invariant guarantees.

References

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